Communication Protocol & Operating System

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WACPv4.0 (Water Automation & Control Protocol)

Architecture: Star Topology (Master + 2 Slaves) Medium: 433MHz RF (FSK Modulation) Bitrate: 9600 bps

Communication Medium

The WACPv4.0 network operates on a high-speed Star Topology. By utilizing FSK Modulation at 9600 bps, the system achieves a 4.8x increase in data throughput compared to the prototype. This increased bandwidth is critical for supporting the new Priority Packet Queue and Double-Tap Fault Reporting.

The network is governed by a Collision Avoidance Strategy using staggered response windows:

- Master (0x10): 250ms backoff (Highest Priority)
- SS (0x02): 500ms backoff (High Priority - Pump/Sump)
- OHS (0x01): 1300ms backoff (Normal Priority - Overhead)

Packet Queuing

To ensure reliable and prioritized packet delivery across the WACP v4.0 network, each station implements an 8-slot outbound packet queue. Packets are inserted into the queue based on a four-tier priority scheme: Priority 1 (Critical) is reserved for fault notifications and amperage reports, Priority 2 (Important) handles pump state changes and timer commands, Priority 3 (Normal) is assigned to periodic level reports and ping responses, and Priority 4 (Low) is used for acknowledgement packets. The queue is sorted at insertion time, ensuring that higher priority packets are always transmitted first regardless of their order of arrival. Each queued entry tracks its retry count and acknowledgement status independently. In the event that the queue reaches full capacity, the lowest priority packet is evicted to accommodate an incoming packet of higher priority — a condition that is considered unlikely under normal operating conditions given the low message frequency of the network. Packets remain in the queue until a valid acknowledgement is received from the destination station, at which point they are removed and the next entry is processed. If no acknowledgement is received within the station's defined backoff period, the packet is retransmitted up to a maximum of three attempts before being discarded gracefully. This queuing mechanism eliminates the risk of critical fault data being lost due to simultaneous transmissions or transient channel interference.

Device Addressing and Identification

Each node is assigned a unique 8-bit hexadecimal address. In the v4.0 architecture, the roles are defined as follows:

- Master Control Unit (0x10): The central coordinator. Manages the 9-step timer, HMI, and global safety locks. It initiates pings and processes telemetry from both satellites.
- Overhead Station (OHS - 0x01): It now monitors the Roof Tank. It uses DHT11-calibrated ultrasonic

sensing to report levels and prevents overflows by enforcing the 80% TANK_HIGH limit.

- Sump & Pump Station (SS - 0x02): The system's "Muscle." It manages P1 (Municipal) and P2 (Transfer). It handles the dual ACS712 multiplexing and manages the isLocked safety states.

Device	Address (HEX)	Description
MASTER	0x10	Central Control Unit (NodeMCU + 4 Buttons)
OHS	0x01	Sump Station (Monitor Only)
SS	0x02	Roof Station (Dual Pump Control + Monitor)

Packet Structure Standard

All communication across the WACP v4.0 network adheres to a fixed 6-byte packet structure to ensure consistency, simplicity, and efficient use of the 9600bps HC-12 radio channel. The first two bytes constitute the preamble, defined as 0xAA followed by 0x55, which serves as a synchronization signature allowing receiving stations to identify the start of a valid transmission. The third byte specifies the destination address, uniquely identifying the intended recipient station — 0x10 for the Master Control, 0x01 for OHS, and 0x02 for SS. The fourth byte carries the command identifier, which instructs the receiving station on how to interpret and act upon the packet. The fifth byte is the value payload, carrying data relevant to the command such as a water level percentage, pump state, fault code, or encoded amperage reading. The sixth and final byte is the checksum, computed as the bitwise XOR of the destination, command, and value bytes, providing a lightweight error detection mechanism to reject corrupted or malformed transmissions. Upon reception, each station validates the preamble, destination address, and checksum before processing the packet — any failure at any of these three checks results in the packet being silently discarded and the receive buffer being flushed. At 9600bps, a single 6-byte packet occupies approximately 5 milliseconds of channel time, making the protocol highly efficient for the low-frequency reporting demands of the system.

Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5
PRE_H	**PRE_L**	**DEST**	**CMD**	**VAL**	**CS**
0xAA	0x55	Target ID	Command	Payload	Checksum

Communication Modalities

The HydroLogix-OS v4.0 implements a **Priority-Stacked Communication Strategy**. This ensures that while the system maintains a constant "Heartbeat," critical safety data always preempts routine telemetry.

1. Timely Communication (The 60s Heartbeat)

To maintain network vitality without clogging the 433MHz frequency, the system uses a 60-second synchronization cycle:

- Master Polling: The Master proactively pings satellites every 60 seconds.
- Slave Silence: If a station has not had a reason to transmit (no level change, no pump change) for 60 seconds, it sends a "Keep-Alive" health report.
- Collision Avoidance: Because the Master (251ms), SS (383ms), and OHS (521ms) use staggered backoff timers, the heartbeat responses are naturally offset, minimizing the risk of packet loss during the sync window.

2. Hybrid Communication (Calibrated Level Monitoring)

Water level data is transmitted using a delta-based logic to preserve bandwidth:

- The 5% Rule: A station only generates a level report if the current water level (compensated by DHT11 temperature data) differs by more than 5% from the last transmitted value.
- Queue Placement: These reports are assigned Priority 3 (NORMAL). If a higher-priority packet (like a Pump Command) is in the 8-slot queue, the level report will wait until the channel is clear.

3. Event-Driven Communication (The "Priority Jump")

Critical changes bypass all timers and are pushed to the top of the 8-slot Priority Queue for immediate broadcast:

- Pump State Changes (Priority 2): When a pump toggles ON or OFF, or when a Timer Preset is updated/expired, the station transmits immediately.
- Fault & Amperage "Double-Tap" (Priority 1): This is the highest priority event in the system.
    1. Step 1: The station stops the pump and sends `CMD_FAULT`.
    2. Step 2: The station immediately follows with `CMD_AMP_Px` containing the trip-current.
    3. Step 3: The Master holds the radio channel open specifically to receive this burst before allowing other traffic.

4. Reliability & The "Stop-and-Wait" Protocol

Regardless of the modality (Heartbeat or Event), every packet requires an ACK (Acknowledge).

- Retry Logic: If an ACK is not received within the backoff window, the station keeps the packet in its Priority Queue and re-attempts the transmission.
- Queue Depth: With 8 slots, the station can buffer multiple events (e.g., a Level Report + a Fault + a State Change) during a brief interference event without losing data.

Control Logic and Safety Standards

The system operates under a Distributed Authority model. While the Master issues "Requests," the satellite stations (specifically SS) perform a local safety audit before execution to ensure hardware integrity and prevent resource wastage.

The Local Safety Audit (The Triple-Check)

Whenever a manual or automated "START" command is received by SS, it must pass three consecutive gates:

- Gate 1: The Lock Check: IF (isLocked == TRUE). If a previous fault (Dry Run/Overload) hasn't been manually cleared by the user, the command is immediately rejected.
- Gate 2: The Level Check: Even if the Master requests a fill, SS will ignore the command if the target tank level is ≥80% (detected via the temperature-calibrated JSN-SR04T). This provides a 20% safety buffer.
- Gate 3: The Watchdog Check: For Pump 2 (P2), the command is only executed if the Disconnect Watchdog is within its 120s heartbeat window.

Pump Identification & State Mapping

To streamline the 8-slot Priority Queue and ensure the Master's UI reflects the real-world state, pump statuses are encoded as follows:

Pump	State	Integer Value	Description
P1	**OFF**	0	Relay open, pump idle.
P1	**ON**	1	Relay closed, municipal draw active.
P1	**LOCKED**	4	**(NEW) Fault detected; requires manual reset.**
P2	**OFF**	2	Relay open, transfer idle.
P2	**ON**	3	Relay closed, C1 to C2 transfer active.
P2	**LOCKED**	5	**(NEW) Fault detected; requires manual reset.**

Command Rejection Feedback

If a command is rejected by a local safety override (e.g., trying to start a locked pump), SS doesn't just stay silent. It pushes a Priority 2 (IMPORTANT) packet back to the Master containing the current LOCKED or OFF state. This forces the Master's UI to "snap back" to the correct state, informing the user that the command was not executed.

Command Structure

The command structure for the WACP Protocol v4.0 is categorized by priority to ensure critical safety data preempts routine telemetry. Priority 1 (Emergency) commands include CMD_FAULT (0x50), which triggers the "Double-Tap" sequence, and the amperage telemetry reports CMD_AMP_P1 (0x40) and CMD_AMP_P2 (0x41), which transmit the trip-current (scaled as VAL × 10). Priority 2 (Action) commands handle the operational logic, featuring CMD_PUMP (0x10) for manual relay control (Value 0-3), CMD_TIMER (0x12) for initiating the 9-step scheduling windows, and CMD_UNLOCK (0x15) for clearing a persistent safety lock. Priority 3 (Telemetry) is reserved for CMD_REPORT_LEVEL (0x30), which carries the temperature-compensated water percentage, and CMD_PING (0x01) for active health polling. Finally, Priority 4 (System) consists of the CMD_ACK (0x02) for packet validation and CMD_RETRY (0x03) for handling checksum failures. This hierarchy ensures that a high-stakes event, such as an overload detected via the 0x40/0x41 commands, jumps to the front of the 8-slot queue to be processed within the Master's 250ms backoff window.

Command Name	HEX	Sender	Payload (VAL) Definition
CMD_PING	0x01	Master	0x00 (Empty) - Requests immediate status update.
CMD_ACK	0x02	All	Returns the CMD ID being acknowledged.
CMD_REPORT	0x31	OHS	Water Level % (0-100).
CMD_REPORT	0x32	SS	Water Level % (0-100).
CMD_PUMP	0x20	Master	0=P1 OFF, 1=P1 ON, 2=P2 OFF, 3=P2 ON.
CMD_PUMP	0x20	SS	Same as above (Reports state change).
CMD_TIMER	0x21	Master	Duration in Minutes (e.g., 30, 60, 90).
CMD_FAULT	0x30	SS	See Fault Code Table below.

Fault Code Definitions

The HydroLogix-OS v4.0 utilizes an expanded 8-bit fault register to distinguish between hydraulic, electrical, and communication failures. By using the ACS712 via the D7 Multiplexer, the system can now identify "Zero-Current" states, which point to hardware breakage (blown fuses or broken wires) rather than just dry pipes. Additionally, the new isLocked safety state is automatically applied to any pump triggering codes 0x01 through 0x06. Unlike the original design, these faults are transmitted via a "Double-Tap" sequence where the Fault Code is immediately followed by a second packet containing the exact trip-amperage, allowing the Master to display precise diagnostic data (e.g., "Overload @ 9.2A") to the user.

Fault Code (HEX)	Fault Name	Trigger Condition	System Action
0x01	**P1_DRY**	P1 Amps < 0.5A	Stop P1, Set isLocked, Send Double-Tap
0x02	**P1_OVER**	P1 Amps > 8.0A	Stop P1, Set isLocked, Send Double-Tap
0x03	**P1_OPEN**	P1 Amps == 0.0A	Alert Master, Set isLocked (Electrical Fail)
0x04	**P2_DRY**	P2 Amps < 0.5A	Stop P2, Set isLocked, Send Double-Tap
0x05	**P2_OVER**	P2 Amps > 8.0A	Stop P2, Set isLocked, Send Double-Tap
0x06	**P2_OPEN**	P2 Amps == 0.0A	Alert Master, Set isLocked (Electrical Fail)
0x07	**COMMS_LOST**	No Master Ping > 120s	SS Freezes P2, OHS enters Low Power
0x08	**C1_EMPTY**	Reservoir Level < 20%	Forced Stop P2 (Interlock)

Latency Characterisation

In the happy path scenario where the queue is empty and no collisions occur, a command travels from Master to SS and receives confirmation in approximately 42ms — 16ms for transmission, 5ms for SS processing, and 16ms for the return ACK. In the realistic worst case where 2–3 lower priority packets are ahead in the queue and one retry is needed due to a collision, end-to-end latency rises to approximately 1.2 seconds, which remains well within acceptable response time for a water automation system. The absolute worst case — where both Master's and SS's queues are fully saturated with 7 competing packets each, every transmission collides and exhausts all 3 retries in both directions, and SS's ultrasonic sensor is mid-read when the packet arrives — produces a theoretical maximum latency of approximately 6.7 seconds; however this scenario requires every possible failure condition to occur simultaneously and is considered statistically negligible in a 3-node system operating on a dedicated HC12 frequency. It is important to note that safety-critical events such as fault reports and emergency stops are assigned PRIORITY_CRITICAL and jump immediately to the front of the queue, guaranteeing delivery confirmation in approximately 37ms regardless of queue state or network conditions.

HydroLogix OS

Hydrologix OS is a lightweight, cooperative, real-time operating system (RTOS) designed specifically for embedded resource management. Unlike general-purpose operating systems (like Windows or Linux) that manage complex files and users, Hydrologix OS is optimized for determinism—ensuring that critical tasks (like stopping a pump during a fault) happen immediately, without delay.

Control Core Architecture

Sequential Polling & Network Synchronization

The Master governs the 433MHz channel through a Staggered Polling Matrix. Instead of flooding the network, it uses a time-sliced heartbeat strategy (pinging OHS at the 1s mark and SS at the 31s mark). This intentional 30-second offset ensures the Master maintains a constant "pulse" on both satellite stations without creating packet collisions, providing a stable foundation for the Star Topology.

Multi-Tiered Time Management

The architecture operates across three distinct temporal resolutions:

- Micro-scale (Real-time): Instantaneous button debouncing and radio packet handling (6-byte buffer monitoring).
- Meso-scale (Minutes): Screensaver timeouts and 30-second countdown updates for the 9-step timer.
- Macro-scale (Daily): NTP Synchronization via Wi-Fi to correct internal crystal drift, ensuring the system clock remains accurate for 24-hour scheduling cycles.

Responsive HMI State Machine

The Master implements an Interrupt-Style UX Logic. Even though it is busy with network coordination, the analog button handler can "wake" the system from its low-power screensaver state immediately. This ensures zero perceived latency for the operator. The UI is decoupled from the automation; it serves as a window into the evaluateAutoLogic() function, which acts as the system's primary decision engine based on the data aggregated from the satellites.

Asynchronous Queue Processing

The Master serves as the network's Data Clearinghouse. By placing processQueue() at the end of every cycle, the OS ensures that high-priority user commands (like a manual Pump Stop) or automatic logic decisions are never blocked by UI rendering or clock updates. Commands are buffered and released only when the channel is clear and the staggered backoff window allows.

In essence, the Master is a Telemetry Aggregator and Policy Enforcer. It takes raw data from the "Eyes" (OHS) and "Muscle" (SS), validates it against the user's manual settings, and issues synchronized directives to keep the entire reservoir system in equilibrium.

OverHead System Core Architecture

1. Environmentally Compensated Sensing: The OS acknowledges that ultrasonic distance is a variable of air temperature. By interleaving a DHT11 thermal check every 10 minutes, the architecture dynamically recalibrates its "Speed of Sound" constant. This ensures that the 80% TANK_HIGH threshold is a physical reality rather than a calculated guess, preventing overflows during high-heat or cold-weather shifts.
2. Synchronous/Asynchronous Hybrid Reporting: The node operates on a "Speak when spoken to, or when things change" logic. It listens for Master polling (Synchronous) but maintains a local Comparator Circuit (Asynchronous) that triggers an immediate broadcast if it detects a level shift greater than 5%. This hybrid approach ensures the Master's UI is always accurate during rapid drainage without spamming the network when the water is static.
3. Radio-Protocol Deference: OHS implements a "Polite Listener" strategy. Before performing the computationally blocking task of reading the 1-wire DHT11 sensor, it checks the HC-12 buffer status. If there is incoming data from the Master, the OS defers the environmental sensing. This ensures that the node never goes "blind" to the network while it is busy calculating temperature, maintaining a 100% response rate to Master Pings.

In essence, OHS is a Deterministic Observer. It is stripped of high-power control logic to focus entirely on Data Integrity and Precision, ensuring that the "Brain" of the system makes decisions based on the most accurate telemetry possible.

Sump Station Core Architecture

Asynchronous Scheduling: The system avoids the "Freeze" typical of simpler controllers. By using internal timestamps instead of delay pauses, the OS can simultaneously track a 30-minute pump timer, monitor real-time amperage, and listen for radio pings without one task slowing down the other.

Channel-Aware Prioritization: The OS is "radio-first." It prioritizes the communication buffer above all else. For instance, it will actually skip a routine temperature reading if the radio channel is busy, ensuring that a "Stop" command from the Master is never missed due to local sensor processing.

Delta-Based Intelligence: To save power and radio bandwidth, the OS is designed to be "quiet." It doesn't report data just for the sake of it; it only triggers a transmission if a significant physical event occurs (like a 5% shift in water level) or if a safety threshold is breached.

Buffered State Management: The logic is decoupled from the transmission. When a fault occurs, the "Brain" of the OS instantly shuts down the hardware and pushes a "Double-Tap" data report into an 8-slot Priority Queue. This ensures that even if the radio environment is noisy, the station won't "forget" to report a fault; it will keep the data in its stack and retry until the Master acknowledges the event.